

Architectural Particular Specification

EXPANSION JOINTS AND MOVEMENT JOINTS

SECTION 6.6

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EXPANSION JOINTS AND MOVEMENT JOINTS

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The work shall consist of propose, supply and installation of all internal and external, fire rated, non-fire-rated, and waterproof expansion joints.

1.02 REFERENCE

A. Section Includes:

1. BS2499:1973 – Specification for Hot Applied Joint Sealants for Concrete pavements.
2. BS2499: Part 2: 1992 – Code of Practice for the Application and Use of Joint Sealants.
3. BS5212: Part 1: 1990 – Specification for Joint Sealants, Cold-Applied for Concrete Pavement.
4. BS5212: Part 2: 1990 – Code of Practice for the Application and Use of Joint Sealants, Cold Applied for Concrete Pavement.
5. BS5889: 1989 – Specification for One-Part Gun Grade Silicone - Based Sealants.
6. BS6093: 1993 – Code of Practice for Design of Joints and Jointing in Building Construction.
7. BS6213: 1982 (1992) – Guide to Selection of Constructional Sealants.
8. BS6262: 1982 – Code of Practice for Glazing for Buildings
9. BS EN 27389: 1991 – Building Construction Jointing Products.
10. BS EN 28340: 1991 – Building Construction Jointing Products. Determination of Tensile properties at Maintained Extension.
11. BS EN 28394: 1991 – Building Construction Jointing Products. Determination of Extrudability of One - Component Sealants.
12. BS EN 29046: 1991 – Building Construction Sealants. Determination of Adhesion/Cohesion Properties at Constant Temperature.

B. American Standard:

1. STM C510 – Test Method for Staining and Color Change of Single- or Multicomponent Joint Sealants.
2. STM C603 – Test Method for Extrusion Rate and Application Life of Elastomeric Sealants.
3. ASTM C639 – Test Method for Rheological (Flow) Properties of Elastomeric Sealant.
4. ASTM C661 – Test Method for Indentation Hardness of Elastomeric-Type Sealants by Mean of a Durometer.
5. ASTM C679 – Test Method for Tack-Free Time of Elastomeric-Sealants.
6. ASTM C717 – Definition of Terms Relating to Building Seals and Sealants.
7. ASTM C719 – Test Method for Adhesion and Cohesion of Elastomeric Joint Sealants Under Cyclic Movement (Hockman Cycle).

8. ASTM C792 – Test Method for Effects of Heat Aging on Weight Loss, Cracking, and Chalking of Elastomeric Sealants.
9. ASTM C793 – Test Method for Effects of Accelerated Weathering on Elastomeric Joint Sealants.
10. ASTM C794 – Test method for Adhesion-in-Peel of Elastomeric Joint Sealants.
11. ASTM C1021 – Laboratory Testing of Sealants.
12. ASTM C1193 – Guide for Use of Elastomeric Joint Sealants.
13. ASTM D2377 – Test Method for Tack-Free Time.
14. ASTM E1399 – Test method for Cyclic Movement Joint Width.

1.03 QUALITY ASSURANCE

(NOT USED)

1.04 PERFORMANCE REQUIREMENTS

- A. Maximum Joint Width: Widest linear gap a joint system tolerates and in which it performs its designed function without damaging its functional capabilities.
- B. Minimum Joint Width: Narrowest linear gap a joint system tolerates and in which it performs its designed function without damaging its functional capabilities.
- C. Movement Capability: Value obtained from the difference between widest and narrowest widths of a joint.
- D. Nominal Joint Width: The width of the linear opening specified in practice and in which the joint system is installed.
- E. Installer Qualifications: Approved by manufacturer and having experience installing joint systems that are similar in design complexity.
- F. Source Limitations: Obtain all architectural joint systems through one source from a single manufacturer.
- G. Product Options: Drawings indicate size, profiles, and dimensional requirements of architectural joint systems and are based on the specific systems indicated. Refer to Division 01 Section "Product Requirements."
 1. Do not modify intended aesthetic effects, as judged solely by Architect, except with Architect's approval. If modifications are proposed, submit comprehensive explanatory data to Architect for review.
- H. Loading Characteristics: Standard loading refers to covers that are capable of withstanding up to 500 lb. point loads. Heavy duty refers to covers that are capable of withstanding up to 2000 lb. point loads.
- I. Fire-Test-Response Characteristics: Where indicated, provide architectural joint system and fire-barrier assemblies identical to those of assemblies tested for fire resistance per UL 2079 and/or ASTM E 1966 by a testing and inspecting agency acceptable to authorities having jurisdiction. Fire rating not less than the rating of adjacent construction.
- J. Manufacturer to provide 5 year warranty for all joint covers.

1.05 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

1.07 SUBMITTALS

A. General

1. Submit manufacturer's specifications and technical data, including material safety data sheets, installation instruction, as required, and catalogue cut and templates where required to explain construction and to provide for incorporation into the project.
2. Submit certificates, copies of independent test reports or research reports shown compliance with fire resistance rating or other specified performance requirements.
3. Product Data: Submit copies of manufacturer's latest published literature for materials specified herein for approval, and obtain approval before materials are fabricated and delivered to the site. Data to clearly indicate movement capability of cover assemblies and suitability of material used in exterior seal for UV exposure.
4. Certificates – Material test reports from qualified independent testing laboratory indicating and interpreting test results relative to compliance of fire-rated expansion joint assemblies with requirements indicated.

B. Shop Drawings

1. Provide shop drawings for each joint system specified and obtain approval prior to fabrication and shipment of materials to the job site.
2. Submit shop drawings showing complete fabrication details for all joint cover, including required anchorage to surrounding construction, recesses, blocking, backing, and connections between similar and dissimilar joint cover assemblies.
3. Provide Placement Drawings that include line diagrams showing plans, elevations, sections, details, splices, block-out requirement, entire route of each joint system, and attachments to other work. Where joint systems change planes, provide isometric or clearly detailed drawing depicting how components interconnect. Alternatively products from a single source supplier are permitted. Confirm compatibility between all elements/materials as appropriate.

C. Samples

1. For purpose of Initial Selection, submit three (3) samples for each type of joint system specified.
2. Include manufacturer's color charts, showing the standard range of colors and finishes available for each exposed metal and elastomeric seal material.

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY, STORAGE AND HANDLING

(NOT USED)

1.10 MOCK-UP

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTINGS

- A. The Contractor shall submit a proposed testing method one month before completion of the installation work to the Architect for review in the proposal. The Contractor shall propose at least one reliable and common testing method for demonstrating the performance of the installed system or material is in accordance with the contract requirement.
- B. The Architect can modify, or delete any test proposed by the Contractor and request the Contractor to perform additional test at the Contractor's expenses if the Contractor's proposal is considered not acceptable.
- C. The Contractor shall arrange the field test work and invite the Architect's site staff to witness the test. Each test shall be recorded and submitted together with photo record in a consolidated report format within one week's time after the test to the Architect for record.

1.12 WARRANTY

This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

The guarantee shall warrant that the entire system(s) including materials, structure, installation, workmanship and performance will be in proper and normal function in accordance with the contract requirements for a period of not less than TEN (10) years from the date of practical completion of the main contract works as certified by the Architect.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 APPROVED MANUFACTURERS

- A. Subject to compliance with the Project Contract Documents requirements, provide joint sealants and required sealant accessories and joint fillers from one or more of the following manufacturers or equal approved by the Architect:

1. Tech Link Construction Engineering Limited
Unit 802, Laford Centre
838 Lai Chi Kok Road
Kowloon
Tel: (852) 3701 1817 / 9316 8906 Fax: (852) 3701 1818
Email: kenuskwong@techlinkcon.com.hk
2. or approved equivalent

2.02 MATERIALS

- A. Material Specification

1. Aluminum: ASTM B 221, Alloy 6063-T5, 6063-T6, 6063-T52, 6061-T5, 6061-T6, 6061-T51, 6105-T5, 6105-T6, 6005-T5, 6005A-T5, 6005A-T61 for extrusions; ASTM B 209, Alloy 6061-T6, 6061-T651, 3003-H14, 5052-H32, 5005-H34 for sheets and plates; 6082-T6, for driveway and vehicular parking areas.
 - a. Apply manufacturer's standard protective coating on aluminum surfaces to be placed in contact with cementitious materials.
 - b. Retain finishes in subparagraphs below to suit Project. Coordinate with Part 2 joint system articles. Verify availability of finishes with manufacturer before specifying. Revise if custom finish is required.
 - c. Mill Finish: AA-M10 (Mechanical Finish: as fabricated, unspecified).
 - d. Select clear or color anodic finish and class from subparagraphs below. Class II is standard finish with many manufacturers. Class I is thicker; verify availability before specifying.
 - e. Class II, Clear Anodic Finish: AA-M12C22A31 (Mechanical Finish: nonspecular as fabricated; Chemical Finish: etched, medium matte; Anodic Coating: Architectural Class II, clear coating 0.010 mm or thicker) complying with AAMA 611.
 - f. Class II, Color Anodic Finish: AA-M12C22A32/A34 (Mechanical Finish: non-specular as fabricated; Chemical Finish: etched, medium matte; Anodic Coating: Architectural Class II, integrally colored or electrolytically deposited color coating 0.010 mm or thicker) complying with AAMA 611.
 - g. Subparagraph below references AAMA standards for high-performance organic coatings on extrusions and panels. Revise if specific products are required.
 - h. High-Performance Organic Finish (Two-Coat Fluoropolymer): AA-C12C40R1x (Chemical Finish: cleaned with inhibited chemicals; Chemical Finish: conversion coating; Organic Coating: manufacturer's standard two-coat, thermocured system consisting of specially formulated inhibitive primer and fluoropolymer color topcoat containing not less than 70 percent polyvinylidene fluoride resin by weight). Prepare, pretreat, and apply coating to exposed metal surfaces to comply with AAMA 2604 and with coating and resin manufacturers' written instructions.

2. Stainless Steel: ASTM A167, A240A, A240M - Type 304 for plates, sheet, and strips.
 - a. Finish: No.4, directional satin.
 - 1) Retain two subparagraphs below if directional finishes are required.
 - 2) Grind and polish surfaces to produce uniform, directionally textured, polished finish indicated, free of cross scratches. Run grain with long dimension of each piece.
 - 3) When polishing is completed, passivate and rinse surfaces. Remove embedded foreign matter and leave surfaces chemically clean.
3. Brass: ASTM B 36/B 36M, UNS Alloy C26000 for half hard sheet and coil.
4. Bronze: ASTM B 455, Alloy C38500 for extrusions; Alloy C28000 Muntz Metal for plates.
5. Elastomeric Seals: Preformed elastomeric membranes or extrusions to be installed in metal frames.
6. Compression Seals: ASTM D2000; preformed rectangular elastomeric extrusions having internal baffle system and designed to function under compression.
7. Fire Barriers: Any material or material combination, when fire tested after cycling, designated to resist the passage of flame and hot gases through a movement joint and to meet performance criteria for required rating period.
8. Moisture Barrier: 7-ply laminate reinforced Polyethylene.
9. Accessories: Manufacturer's standard anchors, clips, fasteners, set screws, spacers, and other accessories compatible with material in contact, as indicated or required for complete installations.
10. PVC Vinyl: Extruded wall and ceiling joint cover.
11. Silicone: Extruded elastomeric flat seal.
12. Extruded Thermoplastic Elastomer: Extruded wall and ceiling joint cover.
13. Polyurethane: Factory installed sealant.
14. Abrasive: Two part epoxy combined with aluminum oxide grit.
15. Water Barrier: Flexible polyvinyl chloride 30 mils, E.D.P.M 45 mils thick or butyl rubber of 3mm CT
16. Fastener, accessories and other materials required for complete installation to manufacturer's instructions.
17. Centering Bars:
 - 4) C-1074 tempered steel with protective coating.
 - 5) Carbon steel.
 - 6) Pantograph bar

B. Tolerance

1. Expansion joint gap throughout development is 50mm.
2. Perpendicular joint movement = $\pm 25\text{mm}$.
3. Parallel to joint movement = $\pm 25\text{mm}$.
4. Vertical/settlement joint movement = 0mm.

- C. Allowable loading on floor joint cover plates and E.J. shall not be less than :
1. 16112.3kg/m² uniform, and 204kg concentrated load for podium internal areas.
 2. 65766kg/m² uniform and 954kg concentrated load for podium driveway pavement, parking and loading/unloading areas and main roofs.

D. Architectural Joint Systems – General

1. Provide architectural joint systems of design, basic profile, materials, and operation indicated. While specified joint systems establish the function and aesthetic intent, it may be necessary for the manufacturer to modify the joint systems to accommodate the movement requirements as scheduled in the contract documents. Such modifications should be made without significant changes to the aesthetic or functional intent of the joint systems. Provide units with capability to accommodate variations in adjacent surfaces.
2. Design architectural joint systems for the following size and movement characteristics:
 - a. Nominal Joint Width: [As indicated on Drawings] [As scheduled] <Insert width>.
 - b. Retain first two subparagraphs below, or delete both and retain third subparagraph.
 - c. Maximum Joint Width: [As indicated on Drawings] [As scheduled] <Insert width>.
 - d. Minimum Joint Width: [As indicated on Drawings] [As scheduled] <Insert width>.
 - e. Lateral Shear Movement Capability: [As indicated on Drawings] [As scheduled] <Insert movement capability>.

E. Architectural Joint Systems – Building Interiors

Internal location as shown in the drawings shall be installed with the following expansion joints/control strips unless otherwise noted.

1. Floor
 - a. Finishes control joint:
Aluminum bearing angle plates and neoprene or synthetic rubber core of minimum. 9mm wide to relieve and isolate stresses in floor finishes.
 - b. Flexible aluminum type expansion joint:
Spring steel centring bar and nylon bearing sphere mechanism with water barrier and block out type anchors.
 - c. Loading & unloading areas and driveways:
Flexible aluminum type expansion joint cover with water barrier and spring steel centring bar, nylon bearing sphere and block out type anchors mechanism to withstand loading as per statutory requirement.
2. Wall
 - a. Aluminum expansion joint cover with water barrier and block out type anchors.
3. Ceiling
 - a. Flexible continuous corrugated profiled thermoplastic rubber expansion strip with mounting side bars/rails.

- b. S/S Gutter shall be provided under the joint

F. Architectural Joint Systems – Exterior Walls and Soffits

External location as shown in the drawings shall be installed with the following expansion joints/control strips unless otherwise noted.

1. Planters

Flexible aluminum type expansion joint cover with water barrier and spring steel centring bar, nylon bearing sphere and block out type anchors mechanism to withstand fire service vehicular lorry and coach loading) above except with water sliding cover plate or membrane profiles for roof detail with resin anchor bolt

2. Covered walkway

Flexible aluminum type expansion joint cover with water barrier and spring steel centring bar, nylon bearing sphere and block out type anchors mechanism except with water sliding cover plate, or membrane profiled for roof detail with resin anchor bolt.

3. External ceiling/ soffits

Flexible continuous corrugated profiled thermoplastic rubber expansion strip with mounting side bars/rails for external use.

4. Vertical cover-plate.

- a. Exposed Metal: aluminum; finish: Class II, clear anodic.
- b. Secondary Seal: 7-ply laminate reinforced Polyethylene.

5. Fire-Resistance Rating

Provide joint system and fire-barrier assembly with a rating not less than that of adjacent construction. (delete if not required)

G. Finishes

- 1. Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.
- 2. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.
- 3. Appearance of Finished Work: Noticeable variations in same piece are not acceptable.
- 4. Aluminum:
 - a. Floors and Roofs: Anodized.
 - b. Wall and Ceilings: Colour anodized to match.
 - c. In contact with concrete: Zinc chromate coated.
- 5. Bronze: Satin finish. [Provide lacquer coating for wall and ceiling joint cover].
- 6. Stainless Steel: satin finish.
- 7. Vinyl: Gray (standard). Option colours to be approved and selected by the Architect.
- 8. Filler strips: Gray (Standard).

H. Allowable loading on floor joint cover plates and E.J. shall not be less than :

- 1. 16112.3kg/m² uniform, and 204kg concentrated load for podium internal areas.

2. 65766kg/m² uniform and 954kg concentrated load for podium driveway pavement, parking and loading/unloading areas and main roofs.

2.03 ANCILLARY MATERIALS

(NOT USED)

PART 3 EXECUTION

3.01 EXAMINATION/INSPECTION

- A. Examine surfaces and block-outs where architectural joint systems will be installed for installation tolerances and other conditions affecting performance of work.
- B. Proceed with installation only after unsatisfactory conditions have been corrected

3.02 PREPARATION

- A. Prepare substrates according to architectural joint system manufacturer's written instructions.
- B. Repair concrete slabs and blockouts using manufacturer's recommended repair grout of compressive strength adequate for anticipated structural loadings.
- C. Coordinate and furnish anchorages, setting drawings, and instructions for installing joint systems. Provide fasteners of metal, type, and size to suit type of construction indicated and to provide for secure attachment of joint systems.

Cast-In Frames: Coordinate and furnish frames to be cast into concrete.

3.03 INSTALLATION/APPLICATION

- A. Fabricate joint cover assemblies as detailed. Provide centring bars, sealing washers, gaskets, splice covers, and closures as necessary for complete installation.
 - 1. Fabricate special transitions and corner fittings as required.
 - 2. Mitre and weld joints as applicable.
 - 3. Provide necessary and related parts, devices, water barrier, anchors, form clips, and other items required for water-barrier installation.
 - 4. Provide corners, tees, transition, curb risers, etc. assembled with connection [mitred] and secured to ensure proper fit and alignment.
 - 5. Special conditions shall be shop fabricated.
- B. Shop assembled components and package with anchor and fittings. Provide components in single lengths where possible; site splicing.
- C. Align work plumb, level, and flush with adjacent surface. Rigidly due to difference between installation and building operating temperatures.
- D. Set Centring bars diagonally at 508mm on centre maximum (or 254mm on centre for heavy duty models). Centring bars shall be fully engaged with the base members.
- E. Water Barrier: Provide water barrier at exterior joints and where called for on drawings. Provide drainage fittings where called for on drawing.
- F. Adjusting and protection:
 - 1. Adjust joint cover to freely accommodate joint movement.
 - 2. Protect installation from damage by work of the other section. Where required, remove and sore cover plate and install temporary protection over joints; re-install cover plate before completion of work.

G. Installation Procedure:

1. Raking out of joint fillers shall be provided by the Contractor for all floor and wall joints at location where E.D.P.M membrane are required to install below the joint covers. Flat and smooth block-out surfaces as required per the approved shop-drawings shall be provided by the Contractor. All patching works shall be of high strength and waterproofed cement mortar.

The block-out surface in the vicinity of the gap shall be cleaned. All dirt, dust, laitance and loose material shall be removed.

2. Fixing:

Prior to lifting the aluminum edge beams into position, check the level and alignment of the joint on site.

Waterproofing membrane shall be installed both on top of and prior to installation of the expansion joint. Contractor should coordinate sequence of works to ensure watertight installation.

The aluminum edge beams shall be secured in position with fixing hardware supplied by the expansion joint manufacturer.

The expansion joint shall then be protected by P.V.C. tape.

3. The void of the recess shall be grouted with high strength waterproof cement and adjacent floor finish shall be supplied immediately.

- H. Comply with manufacturer's written instructions for storing, handling, and installing architectural joint assemblies and materials unless more stringent requirements are indicated.

- I. Metal Frames: Perform cutting, drilling, and fitting required to install joint systems.

1. Install in true alignment and proper relationship to joints and adjoining finished surfaces measured from established lines and levels.
2. Adjust for differences between actual structural gap and nominal design gap due to ambient temperature at time of installation. Notify Architect where discrepancies occur that will affect proper joint installation and performance.
3. Cut and fit ends to accommodate thermal expansion and contraction of metal without buckling of frames.
4. Locate in continuous contact with adjacent surfaces.
5. Standard-Duty Systems: Shim to level where required. Support underside of frames continuously to prevent vertical deflection when in service.
6. Heavy-Duty Systems: Repair or grout blockout as required for continuous frame support and to bring frame to proper level. Shimming is not allowed.
7. Locate anchors at interval recommended by manufacturer, but not less than 3 inches from each end and not more than 24 inches o.c.

- J. Seals in Metal Frames: install elastomeric seals and membranes in frames to comply with manufacturer's written instructions. Install with minimum number of end joints.

1. Provide in continuous lengths for straight sections.
2. Seal transitions according to manufacturer's written instructions. Vulcanize or heat-weld field-spliced joints as recommended by manufacturer.
3. Installation: Mechanically lock seals into frames or adhere to frames with adhesive or pressure-sensitive tape as recommended by manufacturer.

- K. Compression Seals: Apply adhesive or lubricant adhesive as recommended by manufacturer before installing compression seals.
- L. Terminate exposed ends of joint assemblies with field- or factory-fabricated termination devices.
- M. Fire-Resistance-Rated Assemblies: Coordinate installation of architectural joint assembly materials and associated work so complete assemblies comply with assembly performance requirements.
 - 1. Fire Barriers: install fire barriers to provide continuous, uninterrupted fire resistance throughout length of joint, including transitions and field splices.

3.04 TOLERANCE

(NOT USED)

3.05 CLEANING

(NOT USED)

3.06 PROTECTION

- A. Do not remove protective covering until finish work in adjacent areas is complete. When protective covering is removed, clean exposed metal surfaces to comply with manufacturer's written instructions.
- B. Protect the installation from damage by work of other Sections. Where necessary due to heavy construction traffic, remove and properly store cover plates or seals and install temporary protection over joints. Reinstall cover plates or seals prior to Substantial Completion of the Work.

END OF SECTION